polarizer’s axis, the more likely it is to pass through.

In Bohm’s version of EPR, two photons with entangled polarization go flying off in opposite directions from a common source, toward two polarizers. The two polarizers are set to measure polarization along the same axis. Because the photons have entangled polarization, when they arrive at the polarizers, they will always do the same thing—they’ll both either be blocked by the polarizers or pass through them. It doesn’t matter what axis the polarizers are set to: as long as the polarizers’ axes match, a pair of entangled photons will always do the same thing at both of them. And, crucially, it doesn’t matter how far apart the polarizers are either: the two photons will always pass through together or be blocked together, regardless of distance.

And this is just as quantum physics says it should be. The single wave function shared by the two entangled photons guarantees that they will always behave in the same manner when encountering two polarizers with matching axes. But that wave function does not specify what they will do—merely that they will do the same thing.

Now, Einstein’s forced choice, the one he had feared was obscured in the EPR paper itself, comes out in vivid relief. Assuming that nature is local, then the only explanation for the perfectly synchronized long-distance choreography of the entangled photons is that they have a prearranged dance routine, one that they agreed upon before flying off from their common source. But the wave function shared by the entangled photons says nothing about any kind of prearrangement. It just guarantees that the photons will always do the same thing at polarizers with the same settings, that they will be perfectly correlated. Therefore, if nature is local, the wave function is not everything—there must be hidden variables. So either quantum physics is incomplete, or nature is nonlocal. We cannot have both locality and completeness in quantum physics. This is Einstein’s forced choice, the heart of the EPR argument.

Bell toyed with this EPR-Bohm thought experiment, trying to construct a model that would maintain all the predicted results of quantum physics while remaining purely local. “Everything I tried didn’t work,” Bell said. “I began to feel that it very likely couldn’t be done. Then I constructed an impossibility proof.”

Einstein had proven that quantum physics must choose between locality